Multi-Graph Contextual Learning for Multimodal Meme Classification

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ABSTRACT

This work proposes a multi-graph contextual classification framework for multimodal meme analysis that leverages relationships between samples rather than treating them independently. Experiments are conducted on the What's Beneath Misogynous Stereotyping (WBMS) dataset containing meme images and captions annotated for misogynistic content. Text and image features are extracted using BERT and CLIP, respectively, and reduced using Principal Component Analysis (PCA). Three similarity graphs are constructed using cosine similarity based k-nearest neighbors: a text semantic graph, a visual semantic graph, and a class-discriminative graph obtained using Linear Discriminant Analysis (LDA). These graphs are processed using Graph Attention Networks (GAT) to learn contextual node representations which are fused for final classification. The proposed model achieved 94% misogyny category classification on WBMS dataset.

1. Introduction

Many researchers have spent time to address the challenges in multimodal meme analysis and classification. Latest studies have examined techniques that take advantage of both visual and textual modalities to optimize performance in tasks such as hate speech detection, misogyny identification, and offensive content classification in memes Kanwar, Chauhan, Singh et al. (2025) Rehman et al. (2025). All of these methods are majorly founded on pretrained models which includes transformer-based architectures for text and vision language models for images, followed by multimodal fusion techniques to integrate the extracted representations Devlin, Chang, Lee and Toutanova (2019) Radford, Kim, Hallacy et al. (2021) Aggarwal et al. (2026). Although these techniques have shown strong potential in picking up on multimodal semantics, successfully modelling inter-modal relationships and including contextual information across samples remains a major obstacle.

Meme, as a form of online content, has a combination of images and textual information, which is generally used for the conveyance of humor, opinions, or views on societal issues. In most cases, this type of online content is used for promoting misogynistic stereotypes and stories about women Kanwar et al. (2025). The detection of this type of online content is a challenge, as in most cases, the online content is implicit, sarcastic, or context-specific, which is not easy for text-based detection techniques. In most cases, the online content is only understandable in an interdisciplinary context, where the combination of images and textual information is used for conveying a different meaning compared to the individual components, which could be innocuous Rehman et al. (2025).

In this respect, a lot of research has gone into addressing the long-existing challenges in multimodal meme classification and analysis. Over the last few years, several research works have explored how best to integrate both images and text, thus enhancing the efficiency of detecting hate speech, misogyny, and other forms of offensiveness in memes Kanwar et al. (2025); Rehman et al. (2025). These models begin with a pre-trained model, a transformer for text and a vision-language model for images, before moving on to multimodal fusion techniques, where both models are combined Devlin et al. (2019); Radford et al. (2021); Aggarwal et al. (2026). While the results are quite impressive in terms of comprehending the overall meaning of a multimodal combination, it is still a challenge to understand how both models interact and how best to integrate contextual information from other samples.

The aim of this research is to study to check whether the modeling of meme relationships can strengthen multimodal classification performance. To achieve this, the formulated approach starts with extraction of text and image features with BERT and CLIP, as these models can extract deep semantic features from captions and images. The extracted

features are of high dimensions, so Principal Component Analysis (PCA) is used to reduce them to a compact form that can be used for graphical depictions. Three graphs have been constructed based on the similarity of features: a text semantic graph to show text relationships, a visual semantic graph to represent image relationships, and a class discriminative graph using Linear Discriminant Analysis (LDA) to describe class discriminative information. These three graphs are then processed by a Graph Attention Network (GAT) to obtain contextual features by aggregating information from similar samples. The contextual features from all three graphs are then consolidated for meme classification.

(a) Kitchen

(b) Working

(c) Leadership

(d) Shopping

Figure 1: Example memes from the WBMS dataset representing different stereotype categories.

What's Beneath Misogynous Stereotyping (WBMS) dataset is used to carry out experiments for this study. It consists of multimodal memes that are a fusion of text captions and images. The memes are annotated for misogynistic content. Memes belong to classes such as cooking, leadership, shopping and work besides non-mysogynistic samples. The dataset is appropriate for testing multimodal methods of detection and classification of misogyny because it contains various types of misogynistic declarations in the memes.

2. Literature Review

Recent advancements in multimodal meme classification have focused on leveraging both textual and visual information to capture complex semantic relationships present in memes. Existing approaches primarily differ in how they perform classification and whether they incorporate relationships between samples.

K-Nearest Neighbors (KNN) Instance-based learning methods such as k-Nearest Neighbors (kNN) perform classification by measuring similarity between samples in the feature space. In this approach, multimodal features are first extracted using pretrained encoders, and distances between feature vectors are computed using metrics such as cosine similarity or Euclidean distance. The class of a given meme is then determined through majority voting among its nearest neighbors. Extensions such as Kang et al. (2021) integrate kNN into neural frameworks, allowing the model to learn similarity functions dynamically. Despite its simplicity and interpretability, kNN suffers from several practical and methodological constraints.

Limitations: High computational cost during inference due to distance calculations with all samples; strong dependence on feature quality; inability to capture global structural relationships; sensitivity to noise and choice of k.

LLM-Based Classification Recent approaches have leveraged large language models (LLMs) and transformer-based architectures for multimodal classification. Models such as BERT Devlin et al. (2019) generate contextual

embeddings by modeling bidirectional dependencies in text, while CLIP Radford et al. (2021) aligns visual and textual modalities in a shared embedding space. These embeddings are then used for classification through fine-tuned neural networks or prompt-based inference. Works such as Kanwar et al. (2025) and Aggarwal et al. (2026) demonstrate strong performance using such approaches due to their ability to capture deep semantic relationships.

Limitations: Lack of explicit modeling of relationships between samples; primarily operates on independent inputs; high computational and memory requirements; performance sensitive to prompt design and fine-tuning strategy; limited incorporation of dataset-level context.

Graph Attention Networks (GAT) Graph-based learning methods address the limitation of independent sample processing by modeling relationships between memes. In this framework, each meme is represented as a node in a graph, and edges are constructed based on similarity measures such as kNN and cosine similarity. Graph Neural Networks (GNNs) propagate information across nodes, allowing each sample to incorporate contextual information from its neighbors. Graph Attention Networks (GAT) further enhance this process by introducing attention mechanisms that assign adaptive importance weights to neighboring nodes Vrahatis et al. (2024). This enables the model to focus on more relevant connections and learn richer contextual representations. Prior work such as Rehman et al. (2025) demonstrates improved performance using GAT-based multimodal frameworks.

Limitations: Dependence on quality of graph construction; typically limited to a single graph representation; increased computational complexity for large graphs; potential over-smoothing in deep graph layers; difficulty in capturing complementary relationships across modalities when using a single graph.

Research Gap and Motivation Despite these advancements, existing methods exhibit clear limitations. kNN-based approaches are restricted to local similarity, LLM-based models lack explicit relational modeling, and graph-based methods typically rely on a single perspective of similarity. There is limited exploration of frameworks that integrate multiple relational views derived from textual, visual, and class-discriminative features.

To address this limitation, the proposed work introduces a multi-graph contextual learning framework that constructs separate graphs for different modalities and integrates them using attention-based graph learning. This approach enables the model to capture both multimodal semantics and inter-sample dependencies in a unified representation.

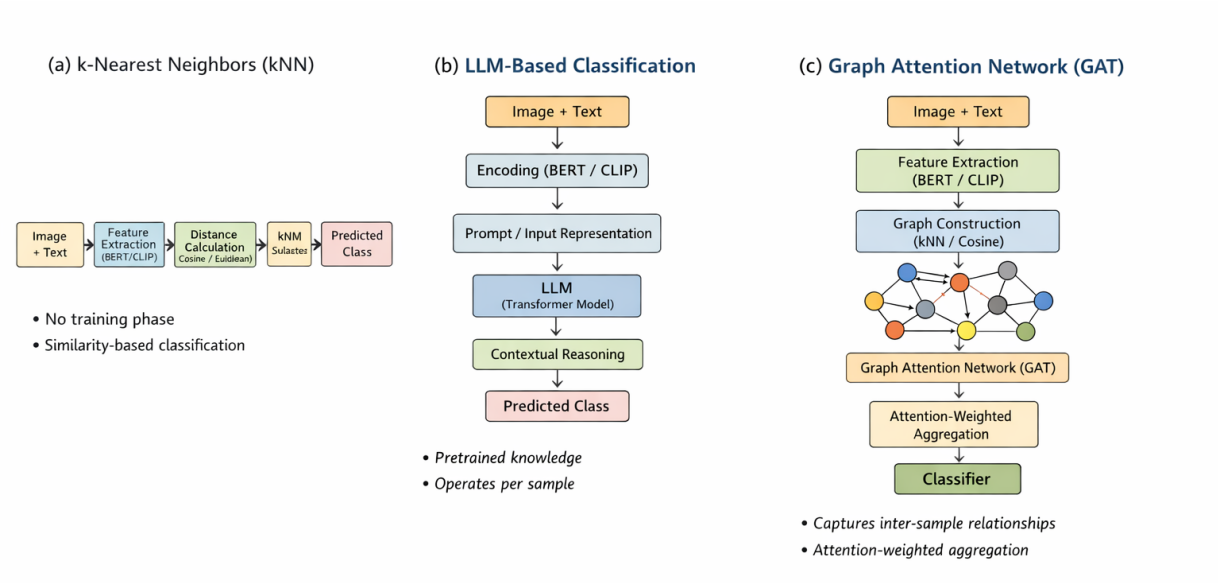

Figure 2: Illustration of existing approaches for multimodal meme classification: (a) kNN, (b) LLM-based classification, and (c) GAT-based learning.

Table 1: Summary of Related Work in Multimodal Meme Classification

S.No.	Title	Year	Authors	Dataset	Methodology	Results	Limitations
1	What is Beneath Misogyny: Misogynous Memes Classification and Explanation	2025	Kanwar et al. (2025)	WBMS	LLM + CLIP multimodal model	F1 score 0.89, better than text-only models	Difficulty detecting implicit bias and humor
2	KNN-GNN: A Powerful Graph Neural Network Enhanced by Aggregating K-Nearest Neighbors in Common Subspace	2024	Li et al. (2024)	Cora, Citeseer	KNN-based GNN	Accuracy: Texas 91.10%, Wisconsin 91.03%, Citeseer 77.14%, Cora 88.62%, Pubmed 88.13%	Extra computation due to KNN search; future work to improve efficiency
3	k-Nearest Neighbor Learning with Graph Neural Networks	2021	Kang et al. (2021)	UCI datasets	Graph-based kNN learning	Error rate example: 0.0838 (ionosphere), 0.0573 (iris); RMSE example: 0.1884 (airfoil), 0.0752 (bikesharing)	Higher computational complexity due to GNN
4	Reasoning Through Memorization: Nearest Neighbor Knowledge Graph Embeddings	2023	Wang et al. (2023)	FB15k-237	kNN-based KGE	WN18RR: Hits@1 0.525, MRR 0.579; FB15k-237: Hits@1 0.280, MRR 0.370	Large knowledge store, higher inference cost
5	A Context-aware Attention and Graph Neural Network-based Multimodal Framework for Misogyny Detection	2025	Rehman et al. (2025)	MAMI, MMHS150K	Multimodal + GNN + Attention	Macro-F1: 87.95% on MAMI, 86.62% on MMHS150K, 10.17% improvement over previous models	High computational complexity; future work could improve efficiency and handle larger multimodal datasets
6	Multimodal Hateful Meme Detection using Knowledge Distillation and Hierarchical Vision Transformer Framework	2026	Aggarwal et al. (2026)	HMC, MMHS150K	KD + ViT	Accuracy: 80.70% (MMHS150K), 62.85% (MultiOff), 87.80% (HMC)	Computational complexity of multimodal transformers; future work can improve efficiency and real-time deployment
7	Application of PCA-Whitening Enhanced BERT for Chinese Semantic Understanding in Short Text Product Name Classification	2024	Zhu et al. (2024)	GS1 China	PCA + BERT	Top-5 accuracy: 95%, Top-1 accuracy up to 99% on high-quality dataset	Performance depends on data quality; future work suggests multi-language, multi-modal models and vector space optimization

3. Dataset

The evaluations in this research are carried out on the What's Beneath Misogynous Stereotyping (WBMS) dataset Kanwar et al. (2025), a multimodal baseline for meme classification. The dataset comprises memes that integrate images and textual material, classified for misogynistic stereotypes. WBMS divides the memes into four classes: Kitchen, Leadership, Working, and Shopping. Every data point may have only an image, or a fusion of image, caption, and overlaid text, appropriate for assessing multimodal learning methods. The dataset has a total of 2130 samples. Table 2 briefly presents the distribution of memes throughout categories.

Table 2: Distribution of samples in the WBMS dataset.

Category	Count	Proportion
Kitchen	1076	0.51
Leadership	534	0.25
Working	321	0.15
Shopping	199	0.09
Total	2130	1.00

The dataset is used to evaluate the effectiveness of the proposed multimodal and graph-based learning framework.

4. Methodology

The proposed methodology is designed as a multi-stage pipeline that systematically integrates multimodal feature extraction with relational learning. Initially, each meme is decomposed into its textual and visual components, from which high-dimensional feature representations are extracted using pretrained language and vision models. To enrich these representations with contextual information, multiple graphs are constructed where nodes correspond to individual memes and edges represent different forms of similarity, including textual semantic similarity, visual feature similarity, and latent topic-level relationships. These graphs provide complementary perspectives on the data distribution and enable the modeling of inter-sample dependencies. Each graph is then processed independently using Graph Attention Networks (GATs), which dynamically assign importance weights to neighboring nodes, allowing the model to learn context-aware embeddings that capture both local and global structure. The resulting embeddings from the different graphs are subsequently integrated through a fusion mechanism to obtain a unified representation that encapsulates multimodal and relational information. Finally, this fused representation is passed through a classification layer to predict the target label. This hierarchical and modular design ensures that the framework effectively leverages both modality-specific features and the underlying relational structure of the dataset.

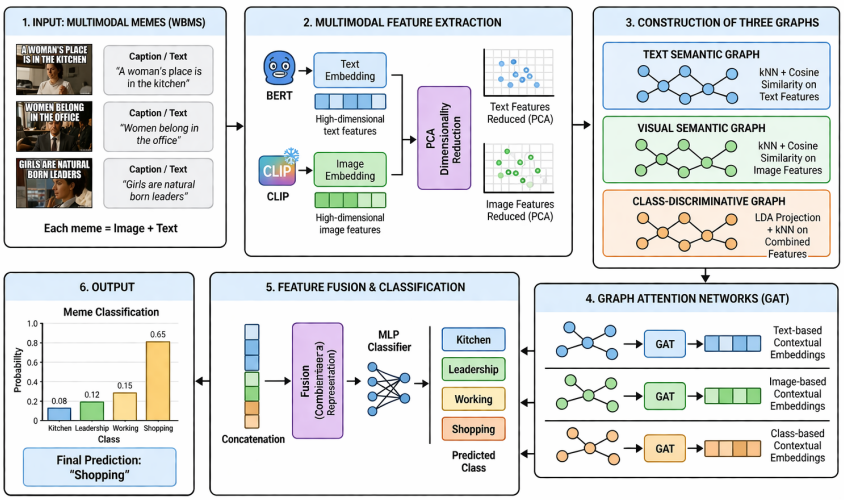

Figure 3: Overview of the proposed multi-graph contextual learning framework.

4.1. Multimodal Feature Extraction

In the first layer, semantic features are extracted from both the textual and visual components of a meme. Textual captions are processed using a BERT model, which generates contextual embeddings that capture how words interact within the given caption. At the same time, the image component is processed using a CLIP-ViT model, which produces visual embeddings that represent how different elements interact within the image. These pretrained models are used because they provide strong and robust feature representations for their respective modalities. Since the extracted embeddings are typically high-dimensional, Principal Component Analysis (PCA) is applied to reduce the dimensionality of both text and image features. This step helps to remove redundant information and produces compact feature representations that are more suitable for subsequent processing.

Principal Component Analysis (PCA): Given a feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, PCA projects it into a lower-dimensional space:

$$\mathbf{Z} = \mathbf{X}\mathbf{W} \tag{1}$$

where $\mathbf{W}$ contains the top-k eigenvectors of the covariance matrix:

$$\mathbf{C} = \frac{1}{n}\mathbf{X}^\top \mathbf{X} \tag{2}$$

4.2. Graph Construction

Using the reduced feature representations obtained from PCA, three k-Nearest Neighbor (kNN) graphs are constructed to model different relationships between memes in the dataset. Each graph captures a distinct type of similarity, enabling the model to incorporate complementary contextual information during representation learning.

Cosine Similarity: The similarity between two feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$ is computed as:

$$\text{sim}(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \tag{3}$$

Text Semantic Graph: A kNN graph is constructed using cosine similarity computed between the PCA-reduced BERT embeddings of meme captions. In this graph, each node represents a meme, and edges connect memes whose textual representations are semantically similar. This structure enables the model to capture linguistic relationships between captions and propagate information across semantically related memes.

k-Nearest Neighbor Graph: An adjacency matrix $\mathbf{A}$ is constructed as:

$$A_{ij} = \begin{cases} 1, & \text{if } \mathbf{x}_j \in \mathcal{N}_k(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

Visual Semantic Graph: A second kNN graph is constructed using cosine similarity between the PCA-reduced CLIP image embeddings. Nodes correspond to meme images, while edges connect visually similar memes. This graph captures visual relationships and shared semantic patterns present in the images.

Class-Discriminative Graph: To incorporate class-level structural information, the multimodal features are projected into a discriminative space using Linear Discriminant Analysis (LDA). A kNN graph is then constructed based on similarities in this projected space. By connecting samples that exhibit similar class characteristics, this graph emphasizes class separability and improves the model's ability to learn discriminative representations.

Linear Discriminant Analysis (LDA): LDA maximizes the ratio of between-class scatter to within-class scatter:

$$\mathbf{W}^* = \arg\max_{\mathbf{W}} \frac{|\mathbf{W}^\top \mathbf{S}_B \mathbf{W}|}{|\mathbf{W}^\top \mathbf{S}_W \mathbf{W}|} \tag{5}$$

4.3. Graph Representation Learning:

The constructed graphs are processed using Graph Attention Networks (GAT). In this framework, each node aggregates information from its neighboring nodes through an attention mechanism that assigns adaptive importance weights. This allows the model to effectively capture contextual dependencies and structural relationships between memes across different graphs.

Graph Attention Network The attention coefficient between node i and node j is computed as:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \,||\, \mathbf{W}\mathbf{h}_j]) \tag{6}$$

The normalized attention weights are obtained using a softmax function:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})} \tag{7}$$

The updated node representation is then computed as:

$$\mathbf{h}'_i = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j\right) \tag{8}$$

4.4. Feature Fusion and Classification

Here, each data point corresponds to a multimodal meme composed of an image and its associated textual content, annotated for misogynistic stereotypes. The fused embedding, obtained by integrating contextual representations from the text, visual, and class-discriminative graphs, is used to classify each meme into one of four categories *Kitchen*, *Leadership*, *Working*, or *Shopping* thereby identifying the specific stereotype conveyed by the meme.

Feature Fusion: The final representation is constructed by concatenating the embeddings learned from each graph:

$$\mathbf{h}_{final} = [\mathbf{h}_{text} \,||\, \mathbf{h}_{image} \,||\, \mathbf{h}_{lda}] \tag{9}$$

This concatenation preserves modality-specific information while enabling the model to learn cross-modal interactions in a unified feature space.

Classification: The fused representation is passed through a Multi-Layer Perceptron (MLP) classifier to produce the final prediction. The output probabilities are computed as:

$$\mathbf{y} = \text{Softmax}(\mathbf{W}_c \mathbf{h}_{final} + \mathbf{b}) \tag{10}$$

Class-Discriminative Analysis with LDA: To evaluate the effectiveness of the class-discriminative graph, we analyze the clustering quality of LDA-projected features using K-Means. Experiments are conducted on text-only, image-only, and combined multimodal representations. The clustering performance is measured using Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI).

Table 3: Clustering evaluation of LDA graphs using different feature sets

Feature Set	**ARI**	**NMI**
Text-only LDA	0.4132	0.4023
Image-only LDA	0.2158	0.2329
Combined Text+Image LDA	0.9269	0.8991

The results demonstrate that the combined multimodal representation significantly improves clustering quality, indicating stronger class separability compared to unimodal features.

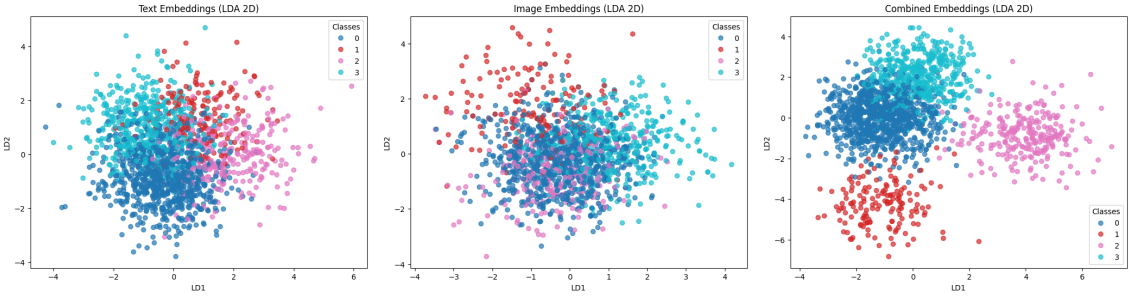

Figure 4: Visual comparison of K-Means clustering on (a) text-only LDA, (b) image-only LDA, and (c) combined multimodal LDA representations

5. Results

The performance of the proposed framework is evaluated across different feature representations and graph-based configurations. For each setup, the model corresponding to the best validation accuracy is selected, and performance is measured using **Accuracy**, **Precision**, **Recall**, **Macro F1-score**, and **Weighted F1-score**.

5.1. Performance Comparison Across Setups

LDA Graph: configuration corresponds to a single-graph model that utilizes only the class-discriminative graph constructed using LDA-projected features. This setup evaluates the effectiveness of learning from discriminative relationships alone.

PCA Two Modalities: configuration does not involve any graph-based learning. It uses PCA-reduced multimodal features (text and image) directly for classification, serving as a feature-only baseline without contextual or relational modeling.

Three-Graph Integration: configuration represents the proposed model, which combines textual, visual, and class-discriminative graphs. This setup captures multiple forms of similarity and leverages Graph Attention Networks to learn contextual representations from all three graphs.

Model	Accuracy	Precision	Recall	Macro F1	Weighted F1
LDA Graph	0.9237	0.8838	0.8702	0.8765	0.9228
PCA Two Modalities	0.9577	0.9266	0.9264	0.9264	0.9577
Three-Graph Integration	0.9331	0.8992	0.8917	0.8951	0.9326

Table 4: Training metrics for different feature and graph setups.

The validation performance for these configurations is reported in Table 5, providing insight into their generalization capabilities.

Model	Accuracy	Precision	Recall	Macro F1	Weighted F1
LDA Graph	0.9484	0.9140	0.9102	0.9106	0.9481
PCA Two Modalities	0.9507	0.9271	0.8965	0.9099	0.9491
Three-Graph Integration	0.9484	0.9003	0.9028	0.9015	0.9485

Table 5: Test (validation) metrics for different feature and graph setups.

5.2. Reference Results on WBMS Dataset

To contextualize the proposed approach, we report the performance of existing methods evaluated on the WBMS dataset. These results are taken from prior work using the MM-Misogyny framework, which formulates the task as multimodal misogyny classification (MMC) along with rationale generation.

It is important to emphasize that the task setting differs from our work. The prior approach focuses on binary classification and explanation generation, whereas our framework addresses multi-class misogyny categorization using graph-based contextual learning. Therefore, the results presented below serve as a reference rather than a direct comparison.

Model	Setting	MMC (F1)
Llama 3	Fine-tuned (Text-only)	0.84
Mistral 7B	Fine-tuned (Text-only)	0.82
OpenHermes 2.5	Fine-tuned (Text-only)	0.80
Llama 3 + CLIP-ViT	Multimodal Fine-tuned	0.89
Mistral 7B + CLIP-ViT	Multimodal Fine-tuned	0.86
OpenHermes 2.5 + CLIP-ViT	Multimodal Fine-tuned	0.85

Table 6: Reference performance of existing methods on the WBMS dataset (reported as MMC F1 score).

The results show that multimodal fine-tuning with visual features (CLIP-ViT) significantly improves performance compared to text-only models. In contrast, our approach explores a different direction by modeling inter-instance relationships through graph structures, enabling richer contextual representation and improved interpretability for multi-class misogyny classification.

5.3. Observations

The LDA-based graph using combined text and image features demonstrates strong validation performance, indicating that class-discriminative projections improve feature separability. The PCA-based two-modality representation achieves the highest training accuracy, showing effective dimensionality reduction, but it does not fully capture relationships between samples. The proposed three-graph integration framework achieves stable and competitive results across all metrics. Although its accuracy is comparable to other setups, it provides better balance between precision, recall, and F1-score, suggesting improved generalization. Overall, the results indicate that incorporating graph-based relational learning alongside multimodal features enhances classification performance, particularly by capturing contextual dependencies between samples.

Despite the promising performance, certain limitations are observed in the proposed framework. The LDA-based graph, while effective for class-discriminative projection, is inherently more suitable for binary or low-dimensional class separation and may not generalize well to more complex multi-class scenarios. This can limit its effectiveness when dealing with nuanced category boundaries. Additionally, the current framework is evaluated on a specific dataset, and its adaptability to more diverse and large-scale datasets remains to be explored.

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